

SLR Protocol

Privacy-Preserving Heart Disease Risk Prediction Using FHE · COMP4101 Practicum · UDST

Oct 9, 2026 · v3 · Abdullah Dar

Review questions

This review asks how fully homomorphic encryption (FHE) has been used to run machine learning inference on health data, and at what cost. Each question maps to a project objective.

RQ	Question	Project objective
RQ1	Which FHE schemes and libraries have been used for ML inference on healthcare data, and why were they chosen?	Encrypted computation
RQ2	Which ML models and approximations of non-linear functions (such as sigmoid) are used, and how does encrypted accuracy compare with plaintext?	Encrypted computation
RQ3	What computational costs are reported, such as latency, ciphertext size, and memory?	Evaluate trade-offs
RQ4	How do studies handle key management, client-side encryption, and fit with clinical workflows?	Key management; practical feasibility

PICOC

The search combines three concepts: encryption, machine learning, and health data. Comparison and outcome terms are used for screening, not in the search string, so that relevant papers are not missed.

Element	Definition for this review	Used in search string?
Population	Healthcare, medical, biomedical, or patient data, such as clinical records, genomic data, or disease prediction datasets	Yes

Element	Definition for this review	Used in search string?
Intervention	Homomorphic encryption (FHE, CKKS, BFV, BGV) applied to ML inference or training	Yes
Comparison	Plaintext ML, or other privacy techniques such as federated learning or secure multi-party computation	No, used in screening
Outcome	Accuracy, latency, ciphertext size, memory, security guarantees	No, used in extraction
Context	ML-as-a-service, cloud or client-server inference	No, used in screening

Search strategy

Three databases are searched, one per pair (see Team roles below): IEEE Xplore (Saif and Naif), ACM Digital Library (Sarim and Hassan), and Scopus (Khalid and Abdullah). The first person named runs the logged search; the second re-runs it the same day to confirm the count. Each search covers papers published 2016 to 2026, in the fields noted under each database below.

Concept	Synonyms (joined with OR)
A. Encryption	"homomorphic encryption", "fully homomorphic", FHE, CKKS, BFV, BGV, TenSEAL, "Microsoft SEAL"
B. Machine learning	"machine learning", "logistic regression", "neural network", "deep learning", "classif*", "predict*", "inference"
C. Health	health*, medical, biomedical, clinical, patient*, disease*, diagnos*, genom*, cancer, diabet*, "heart disease", cardiovascular

Master string: A AND B AND C. Concept C names common diseases (cancer, diabetes) and genomic data because many encrypted-ML papers describe their dataset that way rather than as "health" data, and "biomedical" is not matched by "medical".

IEEE Xplore (Command Search, Abstract field)

```
("Abstract":"homomorphic encryption" OR "Abstract":"fully homomorphic" OR
"Abstract":FHE OR "Abstract":CKKS OR "Abstract":BFV OR "Abstract":BGV OR
"Abstract":TenSEAL OR "Abstract":"Microsoft SEAL") AND ("Abstract":"machine
learning" OR "Abstract":"logistic regression" OR "Abstract":"neural network"
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OR "Abstract": "deep learning" OR "Abstract": classif* OR "Abstract": predict*  
OR "Abstract": inference) AND ("Abstract": health* OR "Abstract": medical OR  
"Abstract": clinical OR "Abstract": patient* OR "Abstract": disease* OR  
"Abstract": biomedical OR "Abstract": diagnos* OR "Abstract": genom* OR  
"Abstract": cancer OR "Abstract": diabet* OR "Abstract": "heart disease" OR  
"Abstract": cardiovascular)
```

Filter: Publication Year 2016 to 2026; Conferences and Journals. This string uses 8 wildcards; IEEE Xplore allows at most 10 per search, so do not add more.

ACM Digital Library (Advanced Search, Edit Query)

```
Abstract: ("homomorphic encryption" OR "fully homomorphic" OR FHE OR CKKS OR  
BFV OR BGV OR TenSEAL OR "Microsoft SEAL") AND Abstract: ("machine learning"  
OR "logistic regression" OR "neural network" OR "deep learning" OR classif*  
OR predict* OR inference) AND Abstract: (health* OR medical OR biomedical OR  
clinical OR patient* OR disease* OR diagnos* OR genom* OR cancer OR diabet*  
OR "heart disease" OR cardiovascular)
```

Filter: Publication date 2016 to 2026; content type Research Article.

Scopus (Advanced document search)

```
TITLE-ABS-KEY ( "homomorphic encryption" OR "fully homomorphic" OR fhe OR  
ckks OR bfv OR bgv OR tenseal OR "Microsoft SEAL" ) AND TITLE-ABS-KEY  
( "machine learning" OR "logistic regression" OR "neural network" OR "deep  
learning" OR classif* OR predict* OR inference ) AND TITLE-ABS-KEY ( health*  
OR medical OR biomedical OR clinical OR patient* OR disease* OR diagnos* OR  
genom* OR cancer OR diabet* OR "heart disease" OR cardiovascular ) AND  
PUBYEAR > 2015 AND PUBYEAR < 2027 AND LANGUAGE ( english ) AND ( DOCTYPE ( ar  
) OR DOCTYPE ( cp ) )
```

IEEE Xplore and ACM search the abstract only, while Scopus searches title, abstract, and keywords.

Abstracts almost always repeat the title's key terms, so this difference is small, but it is reported as a limitation.

Test each string before the logged run, and record the exact string actually used, the date, and the result count.

If any database returns more than about 400 results, narrow concept B to its first four terms ("machine learning", "logistic regression", "neural network", "deep learning") in all three databases, not just that one, so the search stays the same everywhere. Record the change in the search log.

Search check

Before the logged run, check that Scopus finds these known on-topic papers. If any is missing, find out which concept failed to match its title, abstract, or keywords, fix the string, and note the change.

Kim, M., Song, Y., Wang, S., Xia, Y., and Jiang, X. (2018). Secure logistic regression based on homomorphic encryption: design and evaluation. *JMIR Medical Informatics*, 6(2), e19.

Kim, A., Song, Y., Kim, M., Lee, K., and Cheon, J. H. (2018). Logistic regression model training based on the approximate homomorphic encryption. *BMC Medical Genomics*, 11(Suppl 4), 83.

Chen, H., Gilad-Bachrach, R., Han, K., Huang, Z., Jalali, A., Laine, K., and Lauter, K. (2018). Logistic regression over encrypted data from fully homomorphic encryption. BMC Medical Genomics, 11(Suppl 4), 81.

Inclusion and exclusion criteria

A paper is included only if it meets every inclusion criterion and no exclusion criterion. Each exclusion is recorded with its code, which feeds the PRISMA diagram.

Code	Inclusion criteria
I1	Applies homomorphic encryption to ML training or inference
I2	Uses healthcare, medical, or patient data, or a health dataset
I3	Reports experimental results, such as accuracy or runtime
I4	Peer-reviewed journal article or conference paper
I5	Published 2016 to 2026, in English

Code	Exclusion criteria
E1	Uses another privacy technique only (federated learning, differential privacy, MPC) without homomorphic encryption. Hybrid papers that combine homomorphic encryption with another technique are kept.
E2	Not applied to health data
E3	Purely theoretical, with no implementation or results
E4	Survey, review, editorial, poster abstract, or thesis (reviews are kept separately as background reading)
E5	Full text not available through UDST library access
E6	Duplicate or earlier version of an included study (keep the most complete version)

Team roles

All six members work on the review in three fixed pairs. Each pair owns one database, one screening set, and part of the synthesis, so the work is shared evenly and every screening decision is made by two people.

Pair	Members	Search	Screening set	Synthesis
Pair 1	Saif, Naif	IEEE Xplore	Set 1	RQ1 schemes and libraries; RQ4 key management and deployment
Pair 2	Sarim, Hassan	ACM Digital Library	Set 2	RQ3 performance costs
Pair 3	Khalid, Abdullah	Scopus, including the search check	Set 3	RQ2 models, sigmoid approximation, and accuracy

Individual roles on top of the pair work:

- Hassan: de-duplication in RefWorks, splitting records into the three sets, the PRISMA tracking sheet, and the final PRISMA diagram.
- Naif: settles screening disagreements in Pairs 2 and 3, and puts the SLR into each progress report.
- Abdullah: protocol owner (keeps this document up to date) and settles disagreements in Pair 1, since Naif is in it.
- Everyone: full-text reading, data extraction, and quality scoring for the papers that pass their own pair's set, split evenly between the two partners.

Screening and selection process

Screening runs in two stages. Every record is screened by two people independently, which is stronger than double-checking only a sample and reduces bias.

1. Export all records from each database into RefWorks (or Rayyan) and remove duplicates. Hassan logs the count in the PRISMA tracking sheet.
2. Hassan splits the de-duplicated records evenly and at random into Sets 1, 2 and 3, and pastes them into the Screening Log with the pair number.
3. Title and abstract screening: both members of the pair screen every record in their set independently, without seeing each other's decisions. Mark each as include, exclude, or maybe; keep it when unsure.
4. The pair compares decisions. The tracking sheet records the agreement rate. Differences are settled by discussion; if the pair still disagrees, Naif decides (Abdullah for Pair 1).
5. Full-text screening: each pair splits the papers that passed its set between its two members. Each person reads their papers in full, applies criteria I1 to I5 and E1 to E6, and records an exclusion code for each paper removed. The partner checks every exclusion.
6. The person who read an included paper also fills its data extraction row and quality score; the partner spot-checks them.

7. Backward snowballing: each person checks the reference lists of their own included papers for relevant studies the search missed, and adds them to the Screening Log as Snowballing.
8. Hassan checks the PRISMA tracking sheet after each stage and fixes any warnings with the pair concerned.

Quality assessment checklist

Each included paper is scored on five questions: Yes = 1, Partly = 0.5, No = 0. Papers scoring below 2.5 out of 5 are kept but flagged as low quality in the synthesis.

#	Question
Q1	Are the aims and research question clearly stated?
Q2	Are the FHE scheme, library, and encryption parameters reported?
Q3	Is the dataset described and publicly available or clearly sourced?
Q4	Are both accuracy and performance (time or size) measured?
Q5	Is the encrypted result compared against a plaintext baseline?

Data extraction form

For every included paper, record one row in the shared synthesis sheet with these fields. Each field answers one of the review questions.

Field	What to record	Answers
Reference	Authors, year, title, venue, DOI	—
Scheme	CKKS, BFV, BGV, TFHE, or other	RQ1
Library	TenSEAL, SEAL, OpenFHE, HELib, Concrete, or other	RQ1
Parameters	Polynomial modulus degree, scale, security level	RQ1
Model	Logistic regression, neural network, or other	RQ2
Activation approximation	Polynomial degree and input range used for sigmoid or ReLU	RQ2

Field	What to record	Answers
Dataset	Name and size	RQ2
Accuracy	Encrypted vs plaintext accuracy, F1, or AUC	RQ2
Performance	Encryption, inference, and decryption time; ciphertext size; hardware used	RQ3
Key management	Where keys are generated and stored; client-side encryption	RQ4
Deployment and workflow	Architecture, clinical setting, stated barriers	RQ4
Limitations	As stated by the authors	All
Quality score	From the checklist above, out of 5	—

Search log

Fill one row per search as it is run. These numbers become the Identification stage of the PRISMA diagram, so record them exactly and never estimate.

Database	Run by	Date run	Exact string used	Filters applied	Results
IEEE Xplore	Saif, Naif				
ACM Digital Library	Sarim, Hassan				
Scopus	Khalid, Abdullah				
Total before de-duplication					
Duplicates removed	Hassan				
Records to screen					

To do before Progress Report 1 (Sun 11 Oct)

Every member has a task before the report goes in.

- ☐ Everyone: read this protocol; send changes to Abdullah by Sat 10 Oct
- ☐ Khalid and Abdullah: run the Scopus search check, then the Scopus search; fill the search log
- ☐ Saif and Naif: run the IEEE Xplore search and fill the search log
- ☐ Sarim and Hassan: run the ACM search and fill the search log
- ☐ Hassan: export all records to RefWorks, remove duplicates, and enter the counts in the PRISMA tracking sheet
- ☐ Naif: add the protocol and search counts to Progress Report 1